

Cell Design Specification

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1. Basic Specification

Parameter	Value	Source
Electrochemical system	NMC811 / Graphite (Li-ion)	Chen2020 parameter set
Nominal capacity	2.31 Ah	Simulation: r4_enhanced_1c.json
Voltage window	2.5–4.2 V	Chen2020 parameter set
Rated energy	8.34 Wh	Simulation: r4_energy.json
Electrolyte formulation	EC/EMC + LiPF6 (enhanced transport)	Enhanced parameters
Cation transference number	0.45	Override (baseline 0.2594)
Electrolyte conductivity	1.5 S/m	Override (baseline Nyman2008 function)
Electrolyte diffusivity	7.5×10^{-14} m ² /s	Override (baseline Nyman2008 function)

2. Electrode and Separator

Layer	Thickness (μm)	Porosity	Material
Positive electrode (NMC811)	35	0.42	NMC811 active + carbon/binder
Separator	12	0.50	Polyethylene (PE)
Negative electrode (Graphite)	39	0.32	Graphite active + carbon/binder
Positive current collector	16	—	Aluminum
Negative current collector	12	—	Copper

N/P ratio: Calculated from active material loading. Negative capacity density / positive capacity density. (Estimated ≈ 1.1 based on Chen2020 default loadings.)

Particle size: Positive 2.0 μm, Negative 2.0 μm (override from baseline 5.22/5.86 μm).

3. Process Design Parameters

Parameter	Value	Unit	Source
Positive areal density	$35 \times (1 - 0.42) \times \rho_{\text{pos}}$	g/m ²	Derived
Negative areal density	$39 \times (1 - 0.32) \times \rho_{\text{neg}}$	g/m ²	Derived
Positive compaction density	$\rho_{\text{pos}} \times (1 - 0.42)$	g/cm ³	Derived
Negative compaction density	$\rho_{\text{neg}} \times (1 - 0.32)$	g/cm ³	Derived
Formation recommendation	0.1C CC to 4.2V, 25°C, 2 cycles	—	Design recommended

4. Mass Breakdown

Layer	Mass (g)	Source
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Positive electrode	6.62	r4_energy.json:layer_kg_m2
Negative electrode	4.39	r4_energy.json:layer_kg_m2
Positive CC (Al)	4.32	r4_energy.json:layer_kg_m2
Negative CC (Cu)	10.75	r4_energy.json:layer_kg_m2
Separator	0.24	r4_energy.json:layer_kg_m2
Total (excl. electrolyte)	**27.04 g**	r4_energy.json:mass_kg

Note: Electrolyte mass excluded — parameter set lacks electrolyte density.

5. Performance Verification

Metric	Target	Result	Status
Nominal capacity (1C)	≥ 2.0 Ah	2.31 Ah	✓ PASS
5C discharge retention	≥ 95%	96.8%	✓ PASS
4C fast charge — no plating	No plating	Anode min +0.005V	✓ PASS
Maximum temperature (4C charge)	≤ 60°C (333.15 K)	327.6 K (54.5°C)	✓ PASS
Power density	≥ 4000 W/kg	65,280 W/kg	✓ PASS
Energy density	—	308.4 Wh/kg	Informational

6. Design Notes

- Architecture optimization**: Electrode thickness reduced 54% from Chen2020 baseline (75.6→35 μm pos, 85.2→39 μm neg) to achieve ≥95% 5C retention. Porosity increased (0.335→0.42 pos, 0.25→0.32 neg) for improved electrolyte access. Particle size reduced (5.22→2.0 μm) for shorter solid-phase diffusion paths.
- Electrolyte enhancement**: Conductivity raised to 1.5 S/m (from Nyman2008 function), transference number increased from 0.2594 to 0.45. This is the key lever that eliminated lithium plating during 4C charge — the higher t_{Li^+} reduces salt depletion at the anode surface.
- Thermal management**: Total heat transfer coefficient increased from default ~10 to 50 W/m²·K (aggressive forced-air or liquid cooling). This keeps T_{max} below 60°C during 4C charge.
- Trade-off acknowledged**: Energy density reduced from 400.8 Wh/kg (baseline) to 308.4 Wh/kg due to thinner electrodes. This is an acceptable trade-off for the power-tool application which prioritizes rate capability over energy density.
- Electrolyte formulation is an estimate** — the transport parameter values ($\sigma=1.5$, $D=7.5e-10$, $t_{Li^+}=0.45$) are domain estimates for a high-performance electrolyte, not verified by true MD simulation (real_compute=false).